

CE 555: Advanced Design of Structures

Course Code	CE 555
Course Name	Advanced Design of Structures
L-T-P-C	3-0-0-3
Prerequisites	(i) Design of Reinforced Concrete Structures (CE351) or Equivalent; (ii) Design of Steel Structures (CE401) or Equivalent
Intended for	UG, PG (MS, MTech), PhD
Distribution	Specialization Core (MTech in Structural Engineering), Elective (UG, MS, PhD)
Approval	23rd BoA (Apr 2018); 16th Senate (Apr 2018)

Course Contents

- **Module 1:** Design philosophies; Probability distributions and modeling for loading and material strength; Advanced materials and their characteristics. Numerical examples on sampling and acceptance criteria; Reliability of structures. [8 Lectures]
- **Reinforced Concrete Structures:** P-M, M- Φ relationships; P-M-V-T interaction; Strut-and-tie model; Design of deep beam and corbel; Design considerations for slab; Yield line theory; Strip theory; Design of shear walls; Compression field theory for shear design; Design against torsion; Durability design concept; Provisions from Indian Standards and international design guidelines. Development of computer code for interaction curves; Numerical examples on analyses and design of RC structural components using the advanced theories covered in the lecture; Computer simulation of behavior of RC structural components. [12 Lectures]
- **Steel Structures:** Stability design; Torsional buckling (pure, flexural and lateral); Design of beam-columns; Provisions from Indian Standards and international design guidelines. Numerical examples on analyses of steel structures and design of components using the advanced theories covered in the lecture. [8 Lectures]

Text Books

1. N. Subramaniam, **Design of Steel Structures**, Oxford University Press, 2008.
2. N. Subramaniam, **Design of Reinforced Concrete Structures**, Oxford University Press, 2013.

References

1. J.K. Wight, J.G. MacGregor, **Reinforced Concrete: Mechanics and Design**, 6th Edition, Pearson Education, 2016.
2. C.G. Salmon, J.E. Johnson and F.A. Malhas, **Steel Structures: Design and Behavior Emphasizing Load and Resistance Factor Design**, 5th edition, Pearson Education, 2009.
3. T. Pauley, M.J.N. Priestley, **Seismic Design of Reinforced Concrete and Masonry Buildings**, John-Wiley & Sons, 1992.
4. A.H. Nilson, D. Darwin, C.W. Dolan, **Design of Concrete Structures**, 15th edition, McGraw-Hill Education, 2015.
5. D. Menon, S. Pillai, **Reinforced Concrete Design**, 3rd Edition, McGraw Hill Education, 2009.